



8604 - Debris Disks as the Origin of High Altitude Clouds in Exoplanet Atmospheres

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Near-infrared interferometric observations have found that it is common for other solar systems to have large quantities of dust orbiting near the star. Transiting planets that are prime targets for JWST also orbit close to their star, are hence are interacting with these dust disks. Recent estimates have found that close-in planets may accrete this interplanetary dust at such high rates as to significantly change the opacity of the upper atmosphere, possibly giving rise to the ubiquitous high altitude aerosols invoked to explain featureless transmission spectra. Such high altitude clouds due to accretion of dust are well known on Earth, and the close-in exoplanets may accrete at vastly higher rates than Earth, leading to correspondingly higher cloud optical depths.

The goal of this project is twofold: (a) to carry out orbital dynamics simulations of the star-planet-dust system to estimate accretion rates for a broad range of planet, star, and dust parameters, and (b) to implement a dust growth scheme to estimate the particle density and size distribution in the upper atmosphere. The resulting models will be compared to JWST observations of a range of planets, from giants orbiting solar-type stars like HD 209458b and HD 198733b, to sub-Neptunes orbiting low-mass stars like GJ 1214b.

OBSERVING DESCRIPTION